

Gentzen in Review

A Proof Theoretic and Philosophical Overview of Gentzen's Consistency Proof of
Peano Arithmetic

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March 2025

Proof Theory

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“What Gentzen had done was prove the obvious by assuming the doubtful.”

(Ted Chiang, Arrival, 2002)

Abstract

In this paper I consider the 1936 proof by Gerhard Gentzen of the consistency of Peano Arithmetic in the context of Hilbert's Program. While providing a general overview of the proof itself and the methods used within, I will also discuss some philosophical considerations relating to Hilbert's pre- and post-Gödelian goals for the foundations of mathematics. Specifically, the philosophical perspectives of Gentzen himself, Takeuti, Hilbert and Bernays in regard to the finite nature of Gentzen's proof, and more generally, transfinite induction up to the theoretic ordinal number ε_0 .

1. Introduction & Historical Context

One century ago, the German mathematician David Hilbert took part in the momentous debate surrounding the ontology of mathematics. In an attempt to reconcile the schools of Intuitionism, Logicism, and Formalism, Hilbert took inspiration from the works by Frege, Cantor, Dedekind, and others and included the problem of the consistency of arithmetic in his talk at the International Congress of Mathematics in 1900 as one of the most fundamental problems for the 20th century. His goals, which he evolved in collaboration with Bernays, von Neumann, Gentzen, and others, became known as Hilbert's Program. Hilbert assumed that infinitary mathematics was an instrumental extension of finite mathematics and furthermore agreed with his contemporaries that finite mathematics must be consistent. Formalized, Hilbert's Program aimed at proving the consistency of Peano Arithmetic (PA), an infinitary theory, using finite means, ensuring that the results achieved by infinitary mathematics are also provable in finite theories. Such a result would answer Hilbert's interpretation of Intuitionists' concerns that finite mathematics is more secure than infinite mathematics and put an end to logicians preferring finite proofs over infinite ones for the validity of their results. Hilbert asserted that:

“[T]he modes of inference employing the infinite must be replaced generally by finite processes that have precisely the same results (...) That, then, is the purpose of my theory. Its aim is to endow mathematical method with the definitive reliability” (Raatikainen, 2003, p. 162) (Hilbert, 1926, p. 370).

Furthermore, Hilbert was convinced that a finite consistency proof of PA could be found. He wrote:

“In mathematics there is no ignorabimus. On the contrary, we can always answer meaningful questions” (Raatikainen, 2003, p. 168) (Hilbert, 1929, p. 233).

Hilbert interpreted the question of consistency of PA as a meaningful one, and proving this with finite means became a centre piece of Hilbert's Program. Unfortunately for Hilbert, Gödel shattered this very goal in 1931 when he published his Incompleteness Theorems which proved directly that it is impossible for PA to answer all meaningful questions, i.e. to be complete, and to prove its own consistency. Since PA cannot prove its own consistency, even if it were reducible to a finite theory this finite theory could also not prove the consistency of PA. (Gödel, 1931)

Even though Gödel provided the coup de grace for Hilbert's original program, the central ideas remained part of the research in the foundations of mathematics. Gödel's Incompleteness Theorems changed the way logicians were researching consistency results and provided the basis for modern Proof Theory. One of the post-Gödelian researchers attempting to salvage Hilbert's ideas was the German logician Gerhard Gentzen. Gentzen published his consistency proof of PA in 1936 and gave rise to a whole new school of ordinal analysis within Proof Theory. (Paolo Mancosu et al., 2021)

In this paper, I aim at providing an overview of the steps in Gentzen's proof. Starting with providing a general proof plan, followed by a more detailed step-by-step reconstruction of the most important steps of the proof and their justifications. After this, I will discuss the philosophical implications of Gentzen's proof in the context of Hilbert's Program before generally concluding with an optimistic outlook.

2. Overview of the Proof

Gentzen's consistency proof of PA represents a well-developed proof transformation process similar to the normalisation and Cut-Elimination used to prove the consistency of the classical sequent calculus LK. The basic idea of the proof is to show that any proof in PA can be transformed into what Gentzen called a "simple" proof of the same sequent. The process of eliminating the induction rule CJ and Cut instances from a proof is a very involved process and my main source, the book "An Introduction to Proof Theory: Normalization, Cut-Elimination, and Consistency proofs" by Mancosu, Galvan, and Zach (Paolo Mancosu et al., 2021), spends three chapters going into the details of this very procedure. It is explained that one needs to assume transfinite induction along the ordinals up to the proof theoretic ordinal ε_0 . For some intuition, ε_0 is the least ordinal number larger than all finite towers of $\omega^{\omega^{\dots^{\omega}}}$, but it is still a countable ordinal number. Having completed the transformation process from any proof to a simple proof of the same sequent, one can easily show that simple proofs cannot derive falsity, or rather the equivalent consistency statement of not deriving the empty sequent. This yields a consistency proof of PA since it is simply impossible to construct a simple proof in adherence to the rules the sequent calculus PA_S which could yield such inconsistency.

3. Gentzen's Consistency Proof of Peano Arithmetic

Gentzen's proof begins with the fact that PA can be characterized in a sequent calculus PA_S , which fully captures the results of PA. PA_S consists of the classical sequent calculus LK, with the addition of the CJ induction rule and mathematical initial sequents. CJ corresponds to the translation of the induction rule of PA to sequent calculus. In PA, the induction rule is $(F(0) \wedge \forall x(F(x) \rightarrow F(x'))) \rightarrow \forall xF(x)$. In sequent calculus, this rule is the following:

$$\frac{F(a), \Gamma \Rightarrow \Delta, F(a')}{F(0), \Gamma \Rightarrow \Delta, F(t)} \text{ CJ}$$

Furthermore, Gentzen differentiates between logical and mathematical initial sequents, and includes both in his theory. A logical initial sequent refers to the traditional $A \Rightarrow A$ where A is an atomic letter. A mathematical initial sequent has the form $s = t \Rightarrow t = s$ where t and s are terms that may contain free variables.

This construction allows that any proof of PA_S is a proof in PA, and similarly to the consistency proof of LK, proving that the empty sequent is not derivable in this system yields a consistency proof of PA. (Paolo Mancosu et al., 2021, p. 271)

3.1. Cut-Elimination and PA

Unfortunately, a straightforward application of a Cut-Elimination Theorem is impossible in PA_S . Consider the following example illustrating this fact:

$$\frac{\frac{F(a), \Gamma \Rightarrow F(a')}{F(0), \Gamma \Rightarrow F(t)} \text{ CJ} \quad F(t) \Rightarrow \Delta}{F(0), \Gamma \Rightarrow \Delta} \text{ CUT}$$

Observe that there is no guarantee that $F(a') \Rightarrow \Delta$ is derivable if $F(t) \Rightarrow \Delta$ is, which would be a necessary requirement in order to eliminate the Cut rule in this example through an inductive proof such as in the consistency proof of LK (Paolo Mancosu et al., 2021, p. 276). Furthermore, even if it were coincidentally possible to derive $F(t) \Rightarrow \Delta \rightarrow F(a') \Rightarrow \Delta$ somehow, the resulting Cut application would yield the following sequent:

$$\frac{F(a), \Gamma \Rightarrow F(a') \quad F(a') \Rightarrow \Delta}{F(a), \Gamma \Rightarrow \Delta} \text{ CUT}$$

One can plainly see that the sequent $F(a), \Gamma \Rightarrow \Delta$ is not of the right form to apply the induction rule CJ (Paolo Mancosu et al., 2021, p. 276). For these reasons, it is simply impossible to naively apply Cut-Elimination to proofs in PA_S . We require a more sophisticated method that considers CJ carefully and covers these cases. This is exactly the method developed by Gentzen in his 1936 paper. The following sections will go into detail on how this methodology works out.

3.2. Simple Proofs

Similarly to how the consistency of LK was achieved by normalization, Gentzen developed a way to simplify a proof.

Definition: A proof in PA_S is called simple if and only if:

- i) It contains no free variables
- ii) It consists only of atomic formulas
- iii) The only rules applied are weakening, contraction, interchange, and atomic cuts.

Note that specifically operational rules and CJ are not allowed and all formulas in the proof are sentences. (Paolo Mancosu et al., 2021, p. 276).

By straightforward induction on the height of the proof, one can show the following proposition:

Proposition: Every sequent in a simple proof is syntactically valid. (Paolo Mancosu et al., 2021, p. 276)

Note here that the concept of syntactically valid instead of truth is a stronger statement. By using syntactic validity, Gentzen avoids dealing with statements that are true but not provable such as the Gödel sentence.

Gentzen's goal evolves, and he is now concerned with demonstrating that any proof in PA_S can be transformed into a simple proof in PA_S . Specifically, if it can be shown that any proof of the empty sequent can be transformed into a simple proof, then by the previous proposition, this proof cannot exist, since the empty sequent is not syntactically valid. This would constitute the consistency of PA. Note that the empty sequent is syntactically invalid, since there is neither a false formula on the left nor a true formula on the right side of the sequent. So now, the consistency proof of PA depends on proving that any proof in PA_S can be transformed into a simple proof in the same system. (Paolo Mancosu et al., 2021, p. 280)

Be aware that I am glossing over some important details such as but not limited to the initial transformation of proofs to regular proofs, which amounts to a sort of normalization process for eigenvariables to facilitate the simplification process. It represents an intermediate step, and it is quite easy to show that any proof in PA_S can be transformed into a regular proof in PA_S . Therefore, the actual goal of Gentzen's proof is to transform any regular proof into a simple proof. (Paolo Mancosu et al., 2021, p. 282)

3.3. Gentzen's Induction and Cut-Elimination Method

The method for transforming a regular proof into a simple proof is more involved and requires the elimination of all CJ inferences and operational rules in a proof. The process involves three steps that are **successively** and **repeatedly** applied to a proof. These three steps are the following: (Paolo Mancosu et al., 2021, p. 287)

- (1) Replace all CJ inferences with Cut rules
- (2) Remove applications of Weakening
- (3) Remove Cut rule applications

Step (1) is achieved by considering suitable CJ inferences, meaning those where the mentioned term t is a closed term, i.e. does not contain free variables. Gentzen's method always replaces one CJ inference at a time and replaces it with multiple Cuts and introduces copies of sub-proofs. By proceeding with an induction on the maximum number of CJ inferences between a boundary and the end-sequent, any and all

CJ inferences are reduced and then replaced by Cuts. This then yields a CJ-free proof. (Paolo Mancosu et al., 2021, p. 289-292)

Now that all CJ inferences are gone from our proof, we proceed with step (2) to remove instances of Weakening. This is not too difficult, yet necessary for step (3) where we require that a cut-formula is principal. The problem with Weakening is that there could be a cut-formula which descended of a formula introduced by an application of Weakening. To ensure the existence of suitable cuts, we therefore must rid any Weakenings from our proof. For this purpose Gentzen demonstrated that for any proof of an atomic end-sequent $\Gamma \Rightarrow \Delta$ where the end-part, so the part following an application of Cut, contains no free variables (and no application of CJ), there exists a proof without Weakening whose end-sequent is $\Gamma^* \Rightarrow \Delta^*$ where Γ^* and Δ^* are sub-sequents of Γ and Δ . The proof of this result is achieved by induction on the number of inferences in the end-part of the initial proof. (Paolo Mancosu et al., 2021, p. 298)

This result is crucial for Gentzen's proof as any proof of the empty sequent thusly implies the existence of a proof of the empty sequent without Weakening, as the empty sequent does not have any sub-sequents. So, in this case $\Gamma^* = \Gamma = \emptyset$ and $\Delta^* = \Delta = \emptyset$. (Paolo Mancosu et al., 2021, p. 298)

Step (3) is the most important and also the most elaborate step. We now have a proof containing no CJ applications and no Weakening, and we want to remove Cuts from it. The most important consequences of this step are that operational rules within the proof are replaced by applications of Weakening, essentially pushing the boundaries upwards. Furthermore, step (3) will transform some CJ applications that were not suitable into suitable CJ applications, allowing a repetition of step (1) to replace it with Cuts. Additionally, singular cuts are replaced by three cuts where two of them are not of a lower Cut-rank than the original cut-formula. It is not clear whether or not this procedure terminates, and this will need to be shown as an additional result (Paolo Mancosu et al., 2021, p. 294). Simply put, the cuts of the same degree are later removed at a repetition of step (2). Finally, raising the boundary all the way up to the initial sequents of the proof allows us to remove all CJ instances and yields a simple proof (Paolo Mancosu et al., 2021, p. 311).

3.4. Ordinal Notation and Termination

As mentioned above, it remains to show that the procedure explained in Section 3.3 terminates. Gentzen required a well-ordered set which also allows for infinite elements within it to measure the heights of the proofs he was doing induction on. Specifically, Gentzen proved that transfinite induction up to the proof theoretic ordinal ε_0 ($TI(\varepsilon_0)$) is sufficient and necessary to prove that the aforementioned procedure terminates. This then completes the proof of the consistency of PA as any proof of the empty sequent could be transformed into a simple proof of the empty sequent, which is impossible. Furthermore, Gentzen showed that $TI(\varepsilon_0) \vdash \text{Con}(\text{PA}) \wedge \text{PA} \vdash TI(\alpha)$ for each $\alpha < \varepsilon_0$. Since PA could never prove its own consistency according to Gödel's Incompleteness proofs, ε_0 is the least ordinal required to show the consistency of PA. (Paolo Mancosu et al., 2021, p. 347)

Even though the ordinals are the conceptualisation that Gentzen picked, Zach points out that the proof does not depend on ordinal theory, or set theory, and could have been recreated in a different construction of ε_0 . (Paolo Mancosu et al., 2021, p. 321)

In fact, since $\text{PA} \not\vdash TI(\varepsilon_0)$, $TI(\varepsilon_0)$ is the only assumption by Gentzen in his consistency proof of PA. A crucial assumption that I will discuss further in the next section.

4. Philosophical Considerations & Discussion

Initially, Gentzen's proof by itself represents a partial realization of Hilbert's Program, since it relies on $TI(\epsilon_0)$, which cannot be formalized within finite mathematics. Still, it remains close at the heart of Hilbert's goals. One can understand the process of simplification described above as a reduction of PA to a quantifier-free, and finite theory extended by the principle of transfinite induction up to ϵ_0 . This is also surprisingly in line with Gödel's approach of using functionals in his *Dialectica* to show that infinite theories can be reduced to quantifier-free theories as a way to salvage Hilbert's Program (Zach, 2005, p. 27). William Tait and others identify this finite theory to be the system of primitive recursive arithmetic (PRA) (extended by $TI(\epsilon_0)$) (Zach, 2005, p. 27). In this sense, Gentzen reduced PA to a finite system, if one accepts that $TI(\epsilon_0)$ is finitely justified. Note that whether or not $TI(\epsilon_0)$ is finitely justifiable, it is neither provable in PA nor with finite means, and this it does not conflict Gödel's Incompleteness theorems.

4.1. Gentzen's Defence of Transfinite Induction

In his 1936 paper "Die Widerspruchsfreiheit der reinen Zahlentheorie" ("The Consistency of pure number theory"), along with his original proof of the consistency of PA, Gentzen included a defence of the finitism of his method.

Gentzen argued that since $TI(\epsilon_0)$ is well-founded, well-ordered, and adheres to the constructive nature of ordinal numbers, it should be accepted as a form of finite reasoning. He explains that his argument does not rely on any infinite sets, since every ordinal up to ϵ_0 is finitely verifiably well-founded. Therefore, no infinite descending sequences could possibly exist, meaning that the proof he provided only deals with finite objects. He argues that intuitionists should accept ordinal notation based on these reasons (Gerhard Gentzen, 1936). He went so far to defend transfinite induction up to ϵ_0 as "indisputable" on these grounds. (Zach, 2005, p. 23)

Since Gentzen was a formalist, he would accept that ordinals are purely syntactical objects devoid of intrinsic meaning. He would therefore not take any issue with the criticism that without infinities, ordinal numbers would not represent anything real. Gentzen argued that ordinal numbers should be accepted as finite symbolic objects.

Schütte accepted that Gentzen's ordinal numbers, and thusly his consistency proofs, are constructive, yet not strictly finite. Zach furthermore explains that "it is still unclear what the philosophical significance of the technical results is." (Zach, 2005, p.27). To clarify his point, he quotes Solomon Feferman:

"to say that the proof has been carried out by transfinite induction on a certain complicated recursive ordering for some very large ordinal tells us nothing about what constructive principles are involved in the proof of this well-ordering." (Zach, 2005, p. 27)

Gentzen's transfinite induction up to ϵ_0 is thus certainly not as "undisputable" as he thought. At least from his formalist point of view, it seemed intuitive to accept the finite nature of $TI(\epsilon_0)$, even though many modern scholars disagree with his assessment.

Hilbert would also probably not fully accept Gentzen's argument as it stands, since $TI(\epsilon_0)$ can not be formalized using finite means. If we accept the non-controversial claim by Tait that Hilbert's finite mathematics refers to PRA for Π_1^0 sentences (Raatikainen, 2003, p. 160-161), then it's provable that $TI(\epsilon_0)$ can not be formalized within this system. Hilbert originally would have demanded a formalization within PA and also within finite means. Although this might have changed, as I will discuss in the next section.

4.2. Takeuti's Eliminators

Another attempt to demonstrate the finite justification of $TI(\epsilon_0)$ was embarked upon by the Japanese logician Gaisi Takeuti. Zach mentions that "Takeuti attempted to give a finitary justification for the proof theoretic ordinal ϵ_0 " noting that "its philosophical underpinnings are thin." (Zach, 2005, p. 26). The use of

the word “attempted” makes it seem like Zach is not convinced that Takeuti, or anyone for that matter, could succeed in demonstrating the purely finite nature of $TI(\varepsilon_0)$.

In Takeuti’s proof from 1987, he employs the notion of “eliminators” to show that all strictly decreasing sequences of ordinals $\alpha < \varepsilon_0$ are finite only using the fact that the natural numbers are well-ordered (Darnell & Thomas-Bolduc, 2018, p. 7). Allowing a minor correction to the method he employed allows his proof to be considered finitistic from a broader set of perspectives, as demonstrated in Darnell & Thomas-Bolduc, 2018, p. 10-11.

Even with these corrections, the heart of the proof remains the same: Takeuti constructs multiple nested inductions to arrive at his result. And although his method for showing that any $\alpha < \varepsilon_0$ is therefore technically finite, more and more nested inductions are required to demonstrate the finiteness of this α the larger it becomes, and the closer it gets to ε_0 . (Darnell & Thomas-Bolduc, 2018, p. 11)

As Darnell and Thomas-Bolduc point out in their 2018 paper, whether or not one accepts that this method is indeed finite and therefore justifies $TI(\varepsilon_0)$ finitistically depends on the specific perspective of finitism one takes (Darnell & Thomas-Bolduc, 2018, p. 11). If, like many logicians do, one accepts William Tait’s assessment of what Hilbert considered “finite”, then finite methods boil down to those formalizable in primitive recursive arithmetic (Raatikainen, 2003, p. 160-161). Since the nested inductions required for Takeuti’s proof are not primitive recursive, they would not be considered finite in this common interpretation of Hilbert’s Program. (Darnell & Thomas-Bolduc, 2018, p. 11)

However, Darnell and Thomas-Bolduc point to a quote attributed to Post-Gödelian Hilbert and his student Bernays as presented in the 1939 version of “Grundlagen der Mathematik” (Foundations of Mathematics) volume 2:

“Certain methods of finitist mathematics which go beyond recursive number theory (in the original sense) have been discussed already in 7 [of vol. I of the Grundlagen], namely the introduction of functions by nested recursion and the more general induction schema. “(Hilbert & Bernays, 1939, p. 340)

Takeuti interprets Hilbert’s point here as being accepting of methods similar to his as being finite. Darnell and Thomas-Bolduc agree that “if we wished to vindicate Hilbert and Bernays’s assertions that Gentzen’s proof conforms to the finitist standpoint—then we have good reason to think that Takeuti’s proof is fine.” (Darnell & Thomas-Bolduc, 2018, p. 11).

5. Conclusion

In conclusion, Gentzen’s proof of the consistency of PA, as a reduction of an infinite theory to a finite one, remains one of the most important results in the beginnings of modern Proof Theory. His method of ordinal analysis sparked a new field of research wherein the proof theoretic ordinals of systems are investigated. His only assumption which cannot be formalized within the reduced and finite theory remains transfinite induction along the ordinals up to ε_0 . Gentzen argued that this type of induction is natural and “indisputable” (Zach, 2005, p. 23), and provided some arguments in favour of this position. Later Takeuti provided his proof that all $\alpha < \varepsilon_0$ are finite, which did not convince everyone, namely Zach (Zach, 2005, p. 26), even though as Darnell and Thomas-Bolduc point out, Hilbert and Bernays would probably have accepted this proof.

Since the inception of Hilbert’s Program, I consider the work of Gentzen and Takeuti momentous in the assessment of what can be achieved regarding consistency results. In my eyes, Takeuti succeeds in formalizing Gentzen’s proof such that Hilbert would be satisfied, given the Incompleteness Theorems. Such results push the development of modern Proof Theory, and the relationship between it and the nuances of philosophical schools within the foundations of mathematics. It is with great interest and optimism that I consider the future of proof theory and philosophy.

References

- Darnell, E., & Thomas-Bolduc, A. (2018). *Takenti's Well-Ordering Proof: Finitistically Fine?* Philpapers.org. <https://philpapers.org/rec/DARTWP-3>
- Gerhard Gentzen. (1936). Die Widerspruchsfreiheit der reinen Zahlentheorie. *Mathematische Annalen*, 112(1), 493–565. <https://doi.org/10.1007/bf01565428>
- Gödel, K. (1931). Über formal unentscheidbare Sätze der “Principia Mathematica” und Verwandter Systeme I. *Monatshefte Für Mathematik Und Physik*, 38, 173–198.
- Hilbert, D. (1926). “Über das Unendliche”, *Mathematische Annalen* 95,161–190. English translation in J. van Heijenoort (ed.) 1967.
- Hilbert, D. (1929). *Probleme der Grundlegung der Mathematik*, *Mathematische Annalen* 102,1–9. English translation in P. Mancosu (ed.) 1998.
- Hilbert, D., & Bernays, P. (1939). *Grundlagen der Mathematik* (Vol. 2, p. 340). Springer.
- Paolo Mancosu, Galvan, S., & Zach, R. (2021). *An Introduction to Proof Theory : Normalization, Cut-Elimination, and Consistency Proofs*. Oxford University Press.
- Raatikainen, P. (2003). Hilbert's Program Revisited. *Synthese*, 137(1/2), 157–177. <https://doi.org/10.1023/a:1026287119565>
- Zach, R. (2005). Hilbert's Program Then and Now. *Elsevier EBooks*, 411–447. <https://doi.org/10.1016/b978-044451541-4/50014-2>